

Education

Specialist* in Astronomy Sep 2014 – Jul 2020
Lomonosov Moscow State University (MSU)

*Equivalent to a combined bachelor's and master's degree

Selected courses:

- Galactic evolution
- Galaxies at high redshifts (for PhD students)
- Neural networks and genetic algorithms
- Scientific programming in Python
- Big data and real-time algorithms

Postgraduate in Astrophysics 2020–2024*
Lomonosov Moscow State University (MSU)

Dissertation: “Analysis of non-circular gas motions in star-forming regions of nearby galaxies”.

Degree pending dissertation defence.

Grants and Awards

Presidential Grant 2020–2026
Annual award for individual tutors whose students successfully participated in international Olympiads as members of national teams
Sirius, Russia

Undergraduate Olympiad in Space Physics
1st Prize, best overall 2020, Ekaterinburg, Russia

International Olympiad on Astronomy and Astrophysics
Bronze medal 2014, Romania
Bronze medal, best team result 2013, Greece

International Astronomy Olympiad 2013
Silver medal, best team result Lithuania

National Olympiad in Geography
1st Prize, best overall 2014, Orenburg, Russia

National Olympiad in Astronomy
1st Prize, best team result 2014, Novgorod
2nd Prize 2013, Oryol, Russia

National Olympiad in Ecology
2nd Prize 2014, Kazan, Russia
2nd Prize 2013, Orenburg, Russia

National Olympiad (regional round) 2013, 2014
7 prizes (record, both years) Moscow, Russia

Involvement

Jury Member	Volunteer Tutor, Lecturer
· Moscow Astronomy Olympiad (since 2015)	PhysLESch Summer Camp Taught lecture courses:
· National Astronomy Olympiad (since 2016)	· Atmospheric Physics · Fractals in the Universe

Skills

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C++	Pandas	Scipy	English
TOPCAT	PyOpenCL	Astropy	Python
MESA	Tensorflow	Scikit-learn	L ^A T _E X

Publications

(1) Egorov et al, “Star formation in the nearby dwarf galaxy DDO 53: interplay between gas accretion and stellar feedback” Monthly Notices of the Royal Astronomical Society, Volume 508, Issue 2, pp.2650-2667, 2021

(2) Vasilyev K.I, Yarovova A.D and Egorov O.V. “Spectral diagnostics of star-forming regions in the galaxy DDO 53”, Proceedings of the 49-th Annual Conference in Space Physics, p.77-78, Ekaterinburg, 2020. DOI:10.15826/B978-5-7996-2935-9.08

(3) Yarovova et al, “Chemical abundances and small-scale gas kinematics in the local star-forming galaxy NGC 2366” 2025, MNRAS, 541, 3913

Research Experience

Dissertation research Jan 2025 – present
Sternberg Astronomical Institute, MSU, Moscow (independent researcher)

- Currently working with same group on stellar feedback and Lyman continuum leakage in the dwarf star-forming galaxy NGC 2366 [3]

Research Assistant, Engineer Nov 2018 – Dec 2024
Extragalactic Astrophysics research group (Oleg Egorov et al.)
Sternberg Astronomical Institute, MSU, Moscow

- Worked on ML-based image recognition for candidate selection of star-forming rings in UV images.

Stellar feedback and ISM in nearby dwarf galaxies 2018–2020
Sternberg Astronomical Institute, MSU, Moscow

- Worked with long-slit spectra and Fabry-Perot interferometric data cubes obtained on 6-meter BTA and photometric data from 2.5-meter telescope of SAI MSU in order to reveal the origin of star formation of starburst galaxy DDO 53 [Publications: (1),(2)].

Supported by RFBR (project#18-02-00976, PI: Egorov) for 01.2018-12.2020

Creation of the database of the observations of nearby galaxies with Fabry-Perot Interferometer at the 6-m SAO RAS telescope 2020–2024
Sternberg Astronomical Institute, MSU, Moscow

- Worked on decomposition of multicomponent spectral lines using adaptive tessellation and deep learning in order to detect and classify areas of supersonic gas motions and expanding gaseous shells in star forming regions of nearby galaxies.

- Improved the algorithm of automatic detection of areas of supersonic gas motions based on asymmetry of Fabry-Perot panoramic spectroscopy line profiles in the FPI data cubes.

Supported by RSCF (project#19-72-00149, PI: Egorov) for 07.2020-06.2021

Junior Coach, Supervisor and Jury Member

Center for Pedagogical Mastership, Moscow 2016 – present
Russian National Team training camp, St.Petersburg 2017–2019

- Assisted in training and selecting 10 members of Russian national team to compete in International Olympiad on Astronomy and Astrophysics among 50+ candidates.

- Created 15+ theoretical, practical and observational exam problems of high complexity, provided solutions and served as a member of Jury Committee.

- Peer-reviewed 2 papers for Astronomy Education.

- Created and taught 2 year-long courses:
 - Practical problems in Astrophysics
 - Data analysis in Astrophysics (Python)

- Supervised up to 4 groups of 2–5 students in parallel to complete 1-month research projects. Project topics:

- Evolution and dynamics of compact binary stars
- Kinematics of binary open clusters based on GAIA astrometry.
- ML-based classification of astrophysical spectra.
- Chaotic motion in dynamical systems (personal)
- Light curve photometry and imaging calibration (personal)